

Predict–Then–Optimize

A Study of Forecast Uncertainty and Its Operational Impact in Shift-Based Hospital Staffing

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 [https://github.com/asinísusanya/predict-then-optimize](https://github.com/asinिसusanya/predict-then-optimize)

Abstract

Efficient nurse staffing is a critical operational challenge in hospital management, requiring a careful balance between service quality and operational cost under demand uncertainty. In practice, demand forecasts are not an end goal but serve as inputs to downstream staffing optimization models. However, most forecasting approaches are evaluated in isolation using predictive accuracy metrics, without accounting for their impact on operational decisions. This study adopts a Predict–Then–Optimize framework to investigate how demand prediction errors affect nurse staffing decisions and total cost in a shift-based hospital ward operating under realistic constraints inspired by the Sri Lankan healthcare system. A synthetic dataset is generated to simulate daily nurse demand across morning, evening, and night shifts, incorporating both routine operations and high-demand casualty periods that are unobservable to forecasting models. Machine learning models are trained to predict shift-level demand using historical and temporal features, and their forecasts are embedded into a mixed-integer linear staffing optimization model with capacity limits, penalty costs, and a fixed service-level requirement. Staffing decisions are evaluated under four demand scenarios: perfect (oracle) forecasts, machine learning forecasts, noisy forecasts, and systematically drifted forecasts. Results show that lower forecasting error does not necessarily lead to better staffing outcomes. While machine learning and noisy forecasts yield lower apparent staffing costs due to systematic under-prediction and demand smoothing, they conceal latent operational risk. In contrast, systematic demand drift leads to persistent staffing bias and higher total cost, exceeding even the oracle benchmark. The findings demonstrate a fundamental misalignment between prediction accuracy and decision quality in staffing systems. Forecasts must therefore be evaluated based on their downstream operational impact rather than predictive metrics alone. This work highlights the importance of decision-aware evaluation when deploying machine learning models for healthcare workforce planning and provides insights that generalize beyond the specific hospital setting considered.

1 Introduction

Efficient nurse staffing is a critical operational problem in hospital management, directly influencing patient safety, quality of care, and operational cost. Hospitals must ensure that an adequate number of nurses are available across different shifts to meet patient demand, while simultaneously operating under limited human resources and budgetary constraints. Inadequate staffing can lead to increased patient waiting times, compromised care quality, and elevated safety risks, whereas excessive staffing results in unnecessary operational costs.

In recent years, **data-driven approaches**, particularly machine learning (ML), have been increasingly adopted to forecast patient demand and inform staffing decisions. These approaches typically rely on historical data to predict future demand and are evaluated using standard forecasting accuracy metrics such as Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE). However, in practical hospital settings, predictions are not the final objective; instead, they serve as inputs to downstream decision-making processes such as staff scheduling and resource allocation.

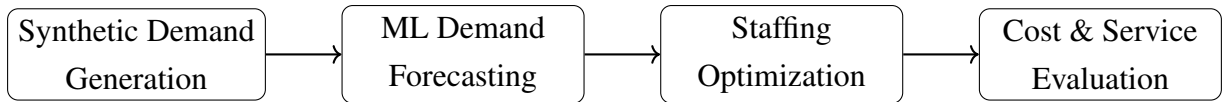


Figure 1: Predict-Then-Optimize framework for nurse staffing

1.1 Motivation

A key limitation of existing demand forecasting approaches is that they are often evaluated **in isolation**, without considering how prediction errors propagate through subsequent optimization or decision models. In practice, staffing decisions are derived by solving an optimization problem that uses forecasted demand as an input. Consequently, even small or seemingly acceptable forecast errors may lead to disproportionately large changes in staffing decisions, service levels, or operational costs.

This disconnect motivates the need for a **Predict-Then-Optimize (PTO)** perspective, where forecasting and optimization are analyzed as a coupled system rather than as independent components. Under this framework, the quality of a forecast is measured not only by its predictive accuracy, but also by its impact on downstream decisions and outcomes.

1.2 Problem Context

This study focuses on a **shift-based nurse staffing problem** in a public hospital ward operating under realistic constraints inspired by the Sri Lankan healthcare system. Staffing decisions are made for three daily shifts morning, evening, and night each with different demand patterns, capacity limits, and operational risks. In particular, night shifts are more sensitive to understaffing due to reduced personnel availability and heightened patient safety concerns.

Patient demand is subject to uncertainty and variability, including occasional high-demand periods such as casualty or emergency days. While such conditions significantly influence staffing requirements, they are often difficult to predict accurately in advance and may not be directly observable by forecasting models.

1.3 Research Question

The central research question addressed in this study is:

How do demand prediction errors affect nurse staffing decisions and total operational cost under realistic hospital constraints?

To answer this question, we explicitly examine how different types of forecast imperfections namely random noise and systematic demand drift impact staffing decisions when forecasts are embedded within an optimization model.

1.4 Objectives

The specific objectives of this study are as follows:

- To simulate realistic, shift-level nurse demand for a hospital ward, incorporating both normal and high-demand (casualty) periods.
- To train machine learning models to forecast nurse demand using historical and temporal features, without access to casualty indicators.
- To formulate a mixed-integer optimization model for nurse staffing that minimizes total cost subject to capacity and service-level constraints.
- To evaluate staffing decisions under different demand scenarios, including perfect forecasts, ML-based forecasts, noisy forecasts, and drifted forecasts.
- To quantify the impact of forecast errors using **decision-aware metrics**, particularly total staffing cost and cost gap relative to an oracle solution.

1.5 Contributions

This work makes the following contributions:

1. It presents a complete Predict–Then–Optimize framework for nurse staffing that integrates machine learning forecasting with operational optimization.
2. It demonstrates that forecast accuracy metrics alone are insufficient to assess staffing quality, as they may not correlate with operational cost.

3. It provides a comparative analysis of random forecast noise versus systematic demand drift, showing that drift poses a significantly higher risk to staffing performance.
4. It offers actionable insights for healthcare workforce planning by highlighting the importance of decision-aware evaluation in predictive systems.

2 System Setup and Assumptions

This study considers a **shift-based nurse staffing problem** for a single public hospital ward operating under realistic constraints inspired by the Sri Lankan healthcare system. This section defines the **operational context**, **staffing structure**, and **modeling assumptions** used throughout the simulation, forecasting, and optimization experiments.

2.1 Hospital and Shift Structure

We consider one hospital ward that operates continuously with **three fixed shifts per day**, consistent with staffing practices in Sri Lankan public hospitals:

Morning 7:00–13:00	Evening 13:00–19:00	Night 19:00–07:00
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Figure 2: Shift structure used for nurse staffing

Let $t = 1, 2, \dots, T$ denote the day index and $s \in \{M, E, N\}$ denote the shift index. Staffing decisions are made independently for each shift on each day, subject to shift-specific constraints.

2.2 Nurse Pool and Capacity Assumptions

Nurses are assumed to be **pooled within each shift**, meaning that all nurses assigned to a given shift are interchangeable and no skill differentiation is considered.

Each shift has a maximum staffing capacity, reflecting the limited number of nurses available for that ward. This is modeled as:

$$x_{t,s} \leq A_s \quad \forall t, s \quad (1)$$

where:

- $x_{t,s}$ is the number of nurses scheduled for shift s on day t
- A_s is the maximum available nurses for shift s

Morning and evening shifts are assumed to have higher capacity than the night shift, reflecting realistic availability patterns.

2.3 Casualty and Non-Casualty Days

Hospital demand is assumed to vary depending on whether a day is a **casualty day** or a **non-casualty day**.

Let:

$$z_t = \begin{cases} 1, & \text{if day } t \text{ is a casualty day} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Casualty days represent periods of unusually high patient inflow due to emergencies or accidents. These days are randomly distributed throughout the planning horizon and are **not included as input features to the machine learning models**. This ensures that forecasting errors naturally arise and reflect realistic uncertainty.

2.4 Demand Characteristics by Shift

Demand for nurses differs systematically across shifts:

- Morning and evening shifts experience higher average demand due to outpatient services, procedures, and admissions.
- Night shifts have lower average demand but are more sensitive to understaffing due to patient safety risks.

These differences are captured through shift-specific demand distributions and penalty costs in the optimization model.

2.5 Cost and Penalty Assumptions

Each nurse assigned to a shift incurs a wage cost, while any unmet demand incurs an understaffing penalty.

Let:

- c_s denote the wage cost per nurse for shift s
- p_s denote the penalty cost per unit of unmet demand for shift s

The following assumptions are made:

- Wage costs are equal for morning and evening shifts.
- Night shift wage cost is higher due to longer duration and workload.
- Understaffing penalties are highest for the night shift, reflecting critical safety concerns.

These assumptions bias the optimization model toward avoiding night shift understaffing even at higher cost.

2.6 Simulation Assumptions Summary

Table 1 summarizes the key assumptions used throughout the simulation, forecasting, and optimization phases.

Table 1: Summary of Simulation and Staffing Assumptions

Component	Assumption
Hospital unit	Single public hospital ward
Planning horizon	12 months (daily resolution)
Shifts per day	3 (Morning, Evening, Night)
Shift structure	Fixed, non-overlapping
Nurse pool	Pooled per shift
Staffing decision	Integer nurses per shift
Capacity limits	Shift-specific maximum availability
Demand variability	Casualty vs non-casualty days
Casualty information	Hidden from ML models
Wage cost	Shift-dependent
Understaffing penalty	Shift-dependent (highest at night)
Service-level requirement	95% minimum coverage
Forecast retraining	Not performed
Objective	Minimize total staffing cost

2.7 Role of Assumptions

The above assumptions serve three purposes:

1. **Realism:** Reflect operational characteristics of Sri Lankan public hospitals.
2. **Controlled experimentation:** Isolate the impact of forecast errors on staffing decisions.
3. **Interpretability:** Ensure observed cost differences arise from prediction–decision interactions rather than arbitrary modeling choices.

3 Data Generation and Exploratory Analysis

This section describes the synthetic nurse demand generation process and presents an exploratory analysis to validate the realism of the simulated data. Since real hospital staffing data is often unavailable due to privacy and operational constraints, a controlled simulation environment is used to generate realistic demand patterns while preserving full observability of the ground truth.

3.1 Synthetic Demand Generation

The planning horizon consists of T consecutive days, representing one year of hospital operations. Each day is divided into three shifts: Morning (M), Evening (E), and Night (N).

Let:

- $t = 1, 2, \dots, T$ denote the day index
- $s \in \{M, E, N\}$ denote the shift index

For each day and shift, the true nurse demand $d_{t,s}$ is generated according to a stochastic process that accounts for both routine operations and high-demand casualty periods.

3.1.1 Casualty Day Assignment

Each day t is classified as either a casualty day or a non-casualty day using a binary indicator:

$$z_t = \begin{cases} 1, & \text{if day } t \text{ is a casualty day} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Casualty days are randomly distributed across the planning horizon and represent periods of unusually high patient inflow due to emergencies or accidents. Importantly, this indicator is used only for data generation and is **not provided to the forecasting models**, ensuring realistic prediction uncertainty.

3.1.2 Shift-Level Demand Model

The true demand for nurses is generated as:

$$d_{t,s} = \mu_s^{(z_t)} + \epsilon_{t,s} \quad (4)$$

where:

- $\mu_s^{(z_t)}$ is the mean demand for shift s under condition z_t
- $\epsilon_{t,s} \sim \mathcal{N}(0, \sigma_s^2)$ is random noise

The mean demand $\mu_s^{(z_t)}$ is higher on casualty days and varies by shift, reflecting differences in patient flow and staffing requirements across the day. The noise term captures unpredictable fluctuations in patient arrivals and operational conditions.

This formulation ensures that demand is:

- Shift-dependent
- Temporally varying
- Stochastic, even under identical conditions

3.2 Generated Dataset

The resulting dataset consists of records of the form:

(date, shift, required_nurses)

Each day contributes exactly three records, one for each shift. This dataset represents the **ground truth demand** and is treated as perfect information only in oracle experiments.

3.3 Exploratory Data Analysis

Exploratory analysis is performed to verify that the simulated demand exhibits realistic and interpretable patterns consistent with hospital operations.

3.3.1 Shift-wise Demand Distribution

Figure 3 shows the distribution of required nurses across shifts.

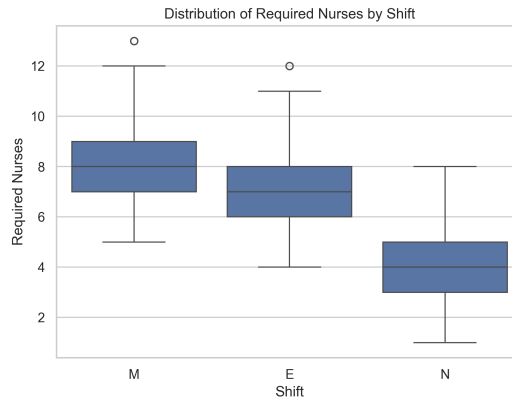


Figure 3: Distribution of required nurses by shift

Interpretation:

- Morning and evening shifts exhibit higher median demand due to routine procedures, admissions, and outpatient services.
- Night shift demand is lower on average but shows noticeable variability.
- These patterns align with expected hospital staffing needs.

3.3.2 Temporal Demand Patterns

Figure 4 illustrates the total daily nurse demand aggregated across all shifts.

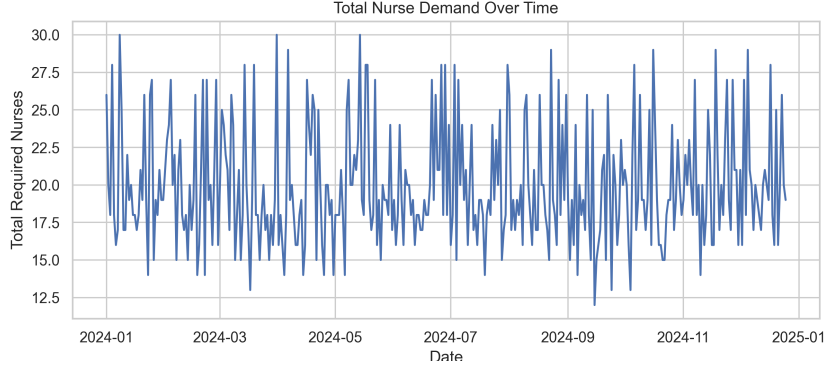


Figure 4: Total nurse demand over time

Interpretation:

- Demand varies over time rather than remaining constant.
- Occasional sharp spikes correspond to simulated casualty days.
- These spikes introduce realistic challenges for forecasting and staffing.

3.4 Role of Exploratory Analysis

The exploratory analysis confirms that the synthetic dataset:

- Exhibits realistic shift-level demand differences
- Contains both routine and high-demand periods
- Presents non-trivial forecasting challenges

These characteristics make the dataset suitable for evaluating how forecast errors propagate into staffing decisions within a Predict–Then–Optimize framework.

4 Demand Forecasting (Machine Learning Layer)

This section describes the machine learning models used to forecast nurse demand and evaluates their predictive performance. The primary purpose of this layer is to generate demand estimates that serve as inputs to the downstream staffing optimization model, rather than to achieve perfect prediction accuracy.

4.1 Forecasting Problem Formulation

For each day t and shift s , the objective of the forecasting model is to estimate the expected nurse demand:

$$\hat{d}_{t,s} = \mathbb{E}[d_{t,s} \mid \mathbf{x}_{t,s}] \quad (5)$$

where:

- $d_{t,s}$ is the true (unknown) nurse demand
- $\hat{d}_{t,s}$ is the predicted demand
- $\mathbf{x}_{t,s}$ is the feature vector available at prediction time

The forecasting task is performed independently for each shift to capture shift-specific demand patterns and variability.

4.2 Feature Engineering

The feature vector $\mathbf{x}_{t,s}$ consists exclusively of historical and temporal information:

$$\mathbf{x}_{t,s} = [\text{dow}_t, d_{t-1,s}, d_{t-7,s}, \text{shift}_s] \quad (6)$$

where:

- dow_t denotes the day-of-week index
- $d_{t-1,s}$ is the demand observed on the previous day
- $d_{t-7,s}$ is the demand observed one week earlier
- shift_s is an encoded representation of the shift

Importantly, the casualty indicator z_t used during data generation is **not included** in the feature set. This ensures that forecasting errors arise naturally from unobserved demand drivers, reflecting real-world uncertainty.

4.3 Forecasting Models

Two widely used regression models are evaluated:

- **Linear Regression (LR):** Serves as a transparent baseline model that captures linear relationships between features and demand.
- **Random Forest (RF):** A non-linear ensemble model capable of capturing complex interactions and non-linear patterns.

Both models are trained separately for each shift using historical data from the training period, and evaluated on a held-out test set.

4.4 Forecast Accuracy Metrics

Forecast performance is evaluated using standard point-wise accuracy metrics.

4.4.1 Mean Absolute Error (MAE)

$$\text{MAE}_s = \frac{1}{N} \sum_t \left| d_{t,s} - \hat{d}_{t,s} \right| \quad (7)$$

4.4.2 Root Mean Squared Error (RMSE)

$$\text{RMSE}_s = \sqrt{\frac{1}{N} \sum_t \left(d_{t,s} - \hat{d}_{t,s} \right)^2} \quad (8)$$

These metrics quantify average prediction error for each shift and model.

4.5 Forecast Accuracy Results

Table 2 summarizes the forecasting accuracy for each shift and model.

Table 2: Forecast accuracy metrics by shift and model

Shift	Model	MAE	RMSE
M	Linear Regression	1.29	1.68
M	Random Forest	1.38	1.76
E	Linear Regression	1.36	1.74
E	Random Forest	1.39	1.78
N	Linear Regression	1.00	1.23
N	Random Forest	1.00	1.24

Figure 5 visualizes the RMSE values across shifts and models.

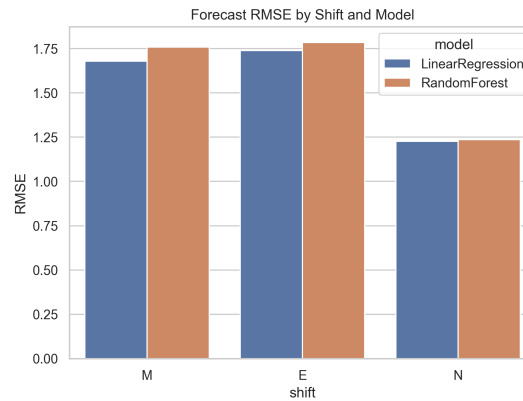


Figure 5: Forecast RMSE by shift and model

Interpretation:

- Forecast accuracy is moderate and similar across both models.

- Random Forest does not significantly outperform Linear Regression.
- Night shift exhibits lower absolute error due to lower demand scale.

These results indicate that increasing model complexity does not necessarily yield substantial accuracy improvements in this setting.

4.6 Forecast Behavior Analysis

To better understand how forecast errors manifest over time, Figures 6a–6c compare true and predicted demand for each shift using the Linear Regression model.

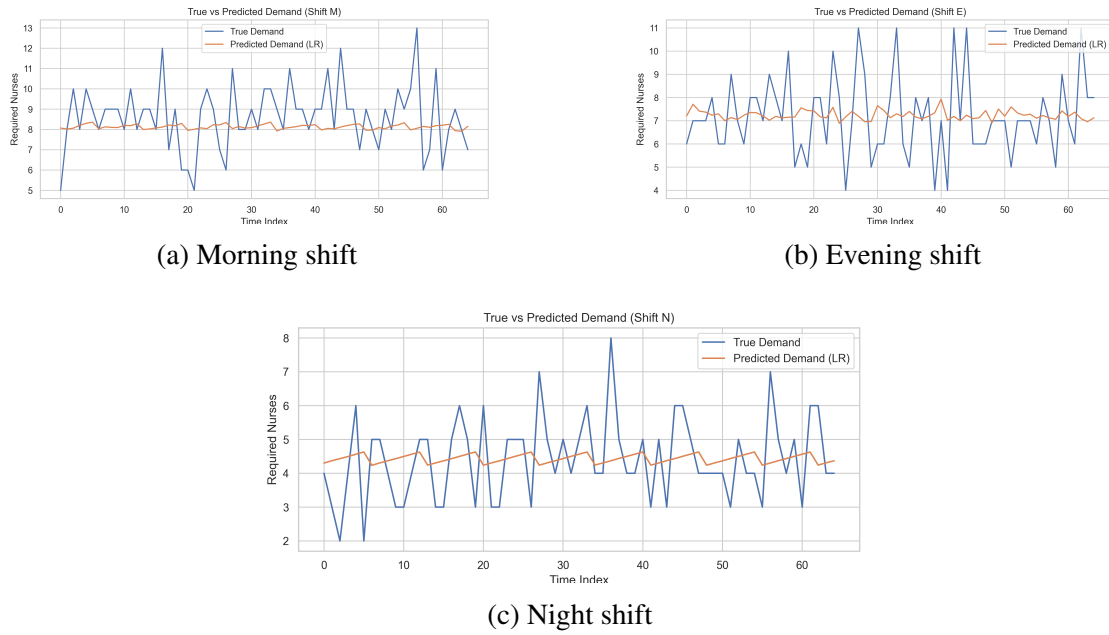


Figure 6: True versus predicted nurse demand across shifts

Key observations:

- Predicted demand tracks overall trends but smooths sharp peaks.
- Extreme demand values, particularly during casualty periods, are under-predicted.
- Forecast bias is systematic rather than random.

This smoothing behavior is particularly important, as staffing decisions are made based on predicted demand rather than true demand.

5 Staffing Optimization Model (Operations Research Layer)

This section presents the optimization model used to derive nurse staffing decisions based on forecasted demand. The model translates predicted demand into concrete staffing actions while accounting for operational constraints, service-level requirements, and cost trade-offs.

The optimization layer constitutes the **decision-making component** of the Predict–Then–Optimize framework, using demand forecasts generated by the machine learning layer as inputs.

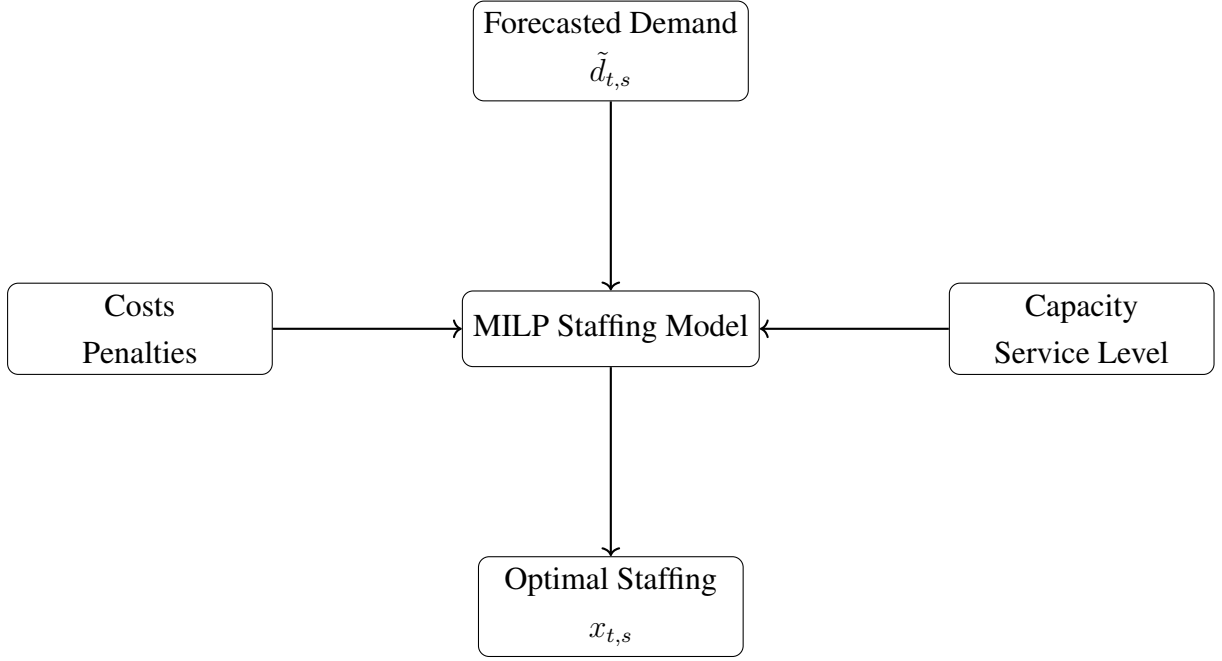


Figure 7: Optimization model for nurse staffing decisions

5.1 Decision Variables

Let:

- $t = 1, 2, \dots, T$ denote the day index
- $s \in \{M, E, N\}$ denote the shift index

The primary decision variable is defined as:

$$x_{t,s} \in \mathbb{Z}_{\geq 0} \quad (9)$$

where $x_{t,s}$ represents the number of nurses assigned to shift s on day t .

To model unmet demand explicitly, an auxiliary slack variable is introduced:

$$u_{t,s} \geq 0 \quad (10)$$

where $u_{t,s}$ represents the amount of unmet (understaffed) demand for shift s on day t .

5.2 Objective Function

The objective of the staffing optimization problem is to minimize the total operational cost over the planning horizon. This cost consists of two components:

- Wage cost incurred by assigning nurses to shifts
- Penalty cost incurred due to understaffing

The objective function is formulated as:

$$\min_{x,u} \sum_{t,s} (c_s x_{t,s} + p_s u_{t,s}) \quad (11)$$

where:

- c_s is the wage cost per nurse for shift s
- p_s is the penalty cost per unit of unmet demand for shift s

Night shifts are assigned higher wage and penalty costs to reflect longer working hours and higher safety risks.

5.3 Demand Coverage Constraint

Let $\tilde{d}_{t,s}$ denote the demand input used by the optimization model. Depending on the experimental scenario, $\tilde{d}_{t,s}$ may correspond to true demand, predicted demand, or perturbed demand. The relationship between staffing, unmet demand, and required demand is enforced by:

$$x_{t,s} + u_{t,s} \geq \tilde{d}_{t,s} \quad \forall t, s \quad (12)$$

This constraint ensures that any shortfall between assigned nurses and demand is captured by the understaffing variable $u_{t,s}$.

5.4 Service-Level Constraint

To enforce minimum service quality, a service-level constraint is imposed:

$$x_{t,s} \geq \alpha \tilde{d}_{t,s} \quad \forall t, s \quad (13)$$

where $\alpha = 0.95$ denotes the required service level.

This constraint ensures that at least 95% of the forecasted demand is met in every shift, preventing extreme understaffing even when forecasts underestimate true demand.

Interpretation: The service-level constraint introduces a conservative bias into the staffing decisions, trading increased cost for improved service reliability.

5.5 Capacity Constraints

Staffing decisions are further constrained by shift-specific nurse availability:

$$x_{t,s} \leq A_s \quad \forall t, s \quad (14)$$

where A_s represents the maximum number of nurses available for shift s .

These constraints reflect practical workforce limitations and prevent unrealistic staffing solutions.

5.6 Integrality and Feasibility

Staffing decisions are required to be integer-valued:

$$x_{t,s} \in \mathbb{Z}_{\geq 0} \quad (15)$$

Because staffing decisions must be integer-valued, the resulting optimization problem is formulated as a Mixed-Integer Linear Program (MILP). The model is implemented using the `PuLP` optimization library in Python and solved with the open-source CBC MILP solver. To ensure feasibility under high demand or limited capacity, controlled understaffing is permitted through non-negative slack variables that incur penalty costs.

5.7 Role of Forecasts in Optimization

The optimization model does not directly observe the true demand $d_{t,s}$, except in oracle experiments. Instead, staffing decisions are derived using forecasted or perturbed demand values:

$$\tilde{d}_{t,s} \in \left\{ d_{t,s}, \hat{d}_{t,s}, \hat{d}_{t,s} + \epsilon, \alpha_s \hat{d}_{t,s} \right\} \quad (16)$$

This design allows direct evaluation of how different forecast qualities influence staffing decisions and total cost.

6 Predict–Then–Optimize Experiments and Results

This section presents the core experimental evaluation of the proposed Predict–Then–Optimize (PTO) framework. The objective is to quantify how different demand forecasting scenarios affect nurse staffing decisions and total operational cost when embedded into the same optimization model.

All experiments use the staffing optimization model described in Section 5 with a fixed service-level requirement of 95%.

6.1 Experimental Setup

Let $d_{t,s}$ denote the true nurse demand and $\hat{d}_{t,s}$ denote the forecasted demand produced by the machine learning model. The optimization model is solved multiple times using different demand inputs $\tilde{d}_{t,s}$ while keeping all other parameters unchanged.

This ensures that any observed difference in staffing cost arises solely from differences in demand estimation rather than changes in the optimization structure.

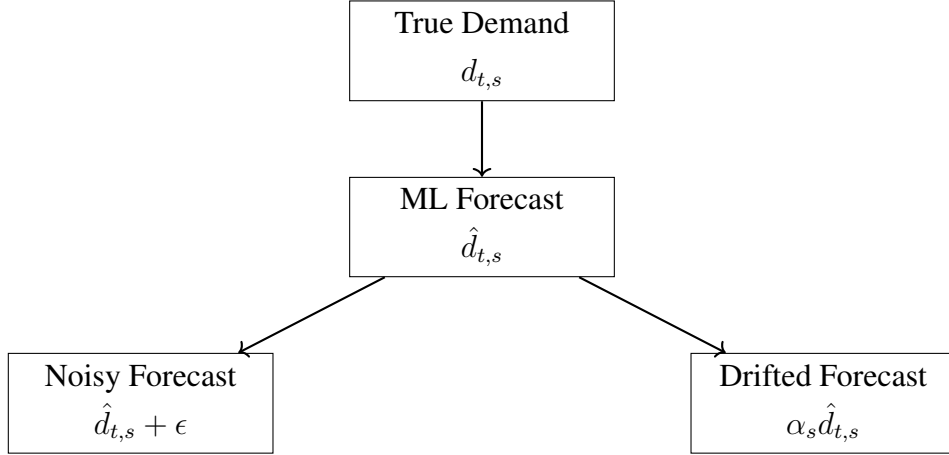


Figure 8: Forecast scenarios evaluated in the experiments

6.2 Evaluated Scenarios

Four demand scenarios are considered:

1. Perfect Forecast (Oracle):

$$\tilde{d}_{t,s} = d_{t,s}$$

This scenario assumes perfect knowledge of future demand and provides a lower bound on achievable staffing cost.

2. ML Forecast:

$$\tilde{d}_{t,s} = \hat{d}_{t,s}$$

Represents realistic deployment where staffing decisions are based on machine learning predictions.

3. Noisy ML Forecast:

$$\tilde{d}_{t,s} = \hat{d}_{t,s} + \epsilon_{t,s}, \quad \epsilon_{t,s} \sim \mathcal{N}(0, 0.1 \hat{d}_{t,s})$$

Models random forecast uncertainty.

4. Drifted ML Forecast:

$$\tilde{d}_{t,s} = \begin{cases} 1.15 \hat{d}_{t,s}, & s \in \{E, N\} \\ \hat{d}_{t,s}, & s = M \end{cases}$$

Models systematic demand drift affecting evening and night shifts.

6.3 Cost Evaluation Metric

For each scenario k , the total staffing cost is computed as:

$$C^{(k)} = \sum_{t,s} \left(c_s x_{t,s}^{(k)} + p_s u_{t,s}^{(k)} \right) \quad (17)$$

To enable decision-aware comparison, the **cost gap** relative to the perfect forecast is defined as:

$$\Delta C^{(k)} = C^{(k)} - C^{(\text{perfect})} \quad (18)$$

This metric captures the operational impact of forecast errors.

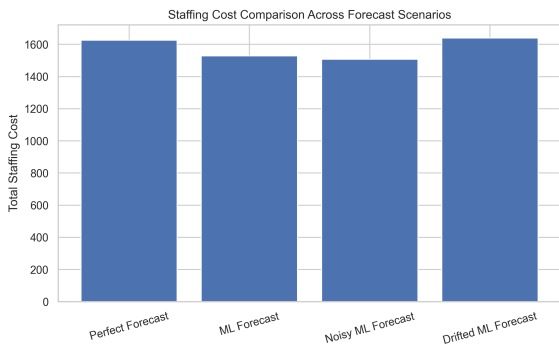
6.4 Cost Results

Table 3 summarizes the total staffing cost and cost gap for all scenarios.

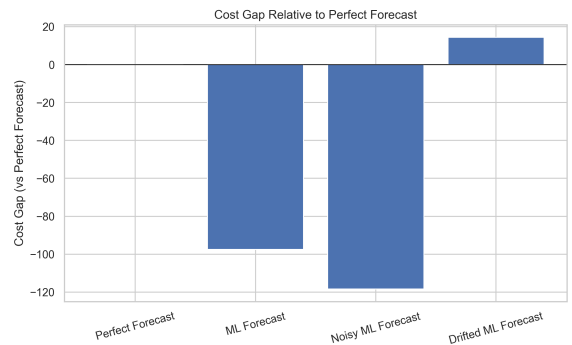
Table 3: Staffing cost comparison under different demand scenarios (95% service level)

Scenario	Service Level	Total Cost	Cost Gap
Perfect Forecast	95%	1625.50	0.00
ML Forecast	95%	1528.01	-97.49
Noisy ML Forecast	95%	1507.08	-118.42
Drifted ML Forecast	95%	1639.85	14.35

6.5 Visualization of Cost Outcomes



(a) Total staffing cost across forecast scenarios



(b) Cost gap relative to perfect forecast

Figure 9: Impact of forecast scenarios on staffing cost

6.6 Results Interpretation

Several important observations emerge from the results:

- The ML-based forecast results in a lower total cost than the perfect forecast, despite being less accurate.
- Introducing random noise further reduces apparent cost by amplifying under-prediction effects.
- In contrast, systematic demand drift increases staffing cost beyond the oracle solution.

These results demonstrate that lower forecasting error does not necessarily lead to better operational outcomes.

6.7 Key Insight

The experimental results reveal a fundamental misalignment between prediction accuracy and decision quality:

$$\text{Lower Forecast Error} \not\Rightarrow \text{Lower Staffing Cost} \quad (19)$$

Random forecast noise may mask operational risk by reducing perceived demand, whereas systematic drift exposes the vulnerability of staffing decisions and leads to cost escalation.

7 Discussion and Insights

This section discusses the implications of the Predict–Then–Optimize experimental results and highlights key insights arising from the interaction between demand forecasting and staffing optimization. Rather than reiterating numerical outcomes, the discussion focuses on explaining why different forecast behaviors lead to distinct operational consequences under the same optimization framework.

7.1 Prediction Accuracy versus Decision Quality

A central finding of this study is the weak relationship between conventional forecasting accuracy metrics and downstream staffing performance. Although the machine learning models achieved reasonable MAE and RMSE values across all shifts, these metrics alone were insufficient to predict staffing cost outcomes.

In multiple scenarios, forecasts with higher prediction error resulted in lower total staffing cost, while more accurate forecasts produced higher costs. This observation reveals a fundamental

misalignment between predictive accuracy and decision quality when forecasts are used as inputs to an optimization model.

Formally, the experimental results demonstrate that:

$$\text{Lower Forecast Error} \not\Rightarrow \text{Lower Staffing Cost}$$

This phenomenon arises because the optimization model reacts nonlinearly to forecast errors due to capacity constraints, penalty structures, and service-level requirements.

7.2 Asymmetric Effects of Forecast Errors

Forecast errors influence staffing decisions asymmetrically depending on their direction.

- **Under-prediction** of demand leads to lower planned staffing levels, reducing wage costs while increasing the likelihood of understaffing penalties.
- **Over-prediction** increases staffing levels, raising wage costs even when true demand is lower.

In the machine learning and noisy forecast scenarios, predicted demand exhibited smoothing behavior that attenuated peak values. This resulted in mild under-prediction, which reduced staffing cost without immediately violating the imposed service-level constraint. While this appears beneficial from a cost perspective, it conceals latent operational risk during high-demand periods.

7.3 Sensitivity of Night Shifts

Across all experiments, night shifts emerged as the most sensitive component of the staffing system. Although average night shift demand is lower than that of morning and evening shifts, the associated understaffing penalties are substantially higher due to patient safety considerations.

As a result, even small forecast deviations during night shifts can have a disproportionate impact on total staffing cost. This sensitivity explains why scenarios involving systematic demand drift affecting evening and night shifts resulted in total costs exceeding those observed under perfect demand information.

7.4 Random Noise versus Systematic Demand Drift

The experiments clearly distinguish between the operational effects of random forecast noise and systematic demand drift.

Random noise introduces short-term variability that tends to average out over time. The optimization model can partially absorb such fluctuations, resulting in limited long-term cost impact.

In contrast, systematic demand drift alters the underlying demand distribution. When forecasts consistently underestimate or overestimate demand, the optimization model repeatedly makes biased staffing decisions. These persistent biases accumulate over time, leading to sustained understaffing or overstaffing and increased operational cost.

The drifted forecast scenario demonstrates that robustness to distributional change is more critical than marginal improvements in point-wise forecasting accuracy.

7.5 Implications of Using a Deterministic Optimization Model

All experiments in this study employ a deterministic mixed-integer optimization model with a fixed service-level constraint. The results show that even under this standard and widely used optimization formulation, forecast imperfections whether random or systematic lead to significant differences in staffing outcomes.

The inclusion of noisy and drifted forecast scenarios demonstrates that meaningful operational risk emerges purely from the interaction between forecast uncertainty and conventional staffing constraints. Consequently, the core Predict–Then–Optimize insights of this study do not rely on advanced robust or stochastic optimization techniques.

While more sophisticated optimization formulations could potentially mitigate some of these effects, they are not required to reveal the fundamental disconnect between prediction accuracy and decision quality highlighted in this work.

7.6 Practical Implications

From a practical standpoint, the findings suggest that hospital staffing systems should not rely solely on forecast accuracy metrics when evaluating predictive models. Instead, forecasts should be assessed based on their downstream impact on staffing decisions, service levels, and operational cost.

In particular:

- Forecast evaluation should be decision-aware rather than purely predictive.
- Night shifts require conservative handling due to their heightened sensitivity to forecast errors.
- Continuous monitoring for demand drift is essential in real-world deployment.

These insights are directly applicable to healthcare workforce planning and extend beyond the specific hospital setting examined in this study.

8 Conclusion and Future Work

This study investigated the interaction between demand forecasting and staffing optimization for a shift-based nurse staffing problem using a Predict–Then–Optimize framework. By integrating machine learning demand forecasts with a mixed-integer optimization model, the study evaluated how different types of forecast imperfections influence staffing decisions and operational cost under realistic hospital constraints.

8.1 Conclusion

The experimental results demonstrate that prediction accuracy alone is insufficient for evaluating the quality of staffing decisions. Although the machine learning models achieved reasonable forecasting performance according to standard accuracy metrics, these metrics did not reliably indicate downstream operational outcomes.

Specifically, the results showed that:

- Machine learning forecasts led to lower apparent staffing cost due to systematic smoothing of demand peaks.
- Random forecast noise further reduced cost by amplifying under-prediction effects, thereby masking latent operational risk.
- Systematic demand drift caused staffing cost to exceed even the oracle benchmark, highlighting the vulnerability of staffing decisions to unobserved distributional changes.

These findings reveal a fundamental misalignment between forecast accuracy and decision quality. Forecast errors interact with service-level requirements, capacity constraints, and penalty structures in non-linear ways, producing outcomes that cannot be inferred from predictive metrics alone.

Overall, this work demonstrates the necessity of decision-aware evaluation when deploying machine learning models in healthcare workforce planning. Even under a standard deterministic optimization framework, forecast imperfections can lead to significant and counterintuitive operational consequences.

8.2 Implications

From a practical perspective, the results suggest that hospital administrators and system designers should exercise caution when relying solely on forecasting accuracy to guide staffing decisions. Instead, predictive models should be evaluated based on their impact on staffing policies, service reliability, and cost.

In particular, the study highlights the importance of:

- Evaluating forecasts within the context of the decisions they support.

- Paying special attention to night shift staffing due to its heightened sensitivity to forecast errors.
- Monitoring for systematic demand drift during long-term deployment of predictive systems.

These insights are directly applicable to real-world hospital operations and extend beyond the specific setting considered in this study.

8.3 Future Work

Several directions for future research can build upon the findings of this work:

- **Robust and stochastic optimization:** Incorporating robust or stochastic optimization techniques to explicitly hedge against forecast uncertainty and demand drift.
- **Adaptive forecasting:** Developing mechanisms for detecting demand drift and retraining forecasting models dynamically.
- **Multi-ward coordination:** Extending the framework to jointly optimize staffing across multiple hospital wards with shared nurse pools.
- **Additional operational constraints:** Modeling nurse fatigue, overtime, skill heterogeneity, and regulatory constraints.
- **Real-world validation:** Applying the framework to real hospital demand data to further validate its practical relevance.

These extensions would enhance the realism of the model while preserving the decision-aware evaluation philosophy emphasized in this study.

Final Remark

This work underscores that the value of predictive models lies not only in their accuracy, but in how they influence real operational decisions. By explicitly linking forecasting and optimization, the Predict–Then–Optimize framework provides a principled approach for evaluating and improving data-driven decision systems in healthcare and beyond.